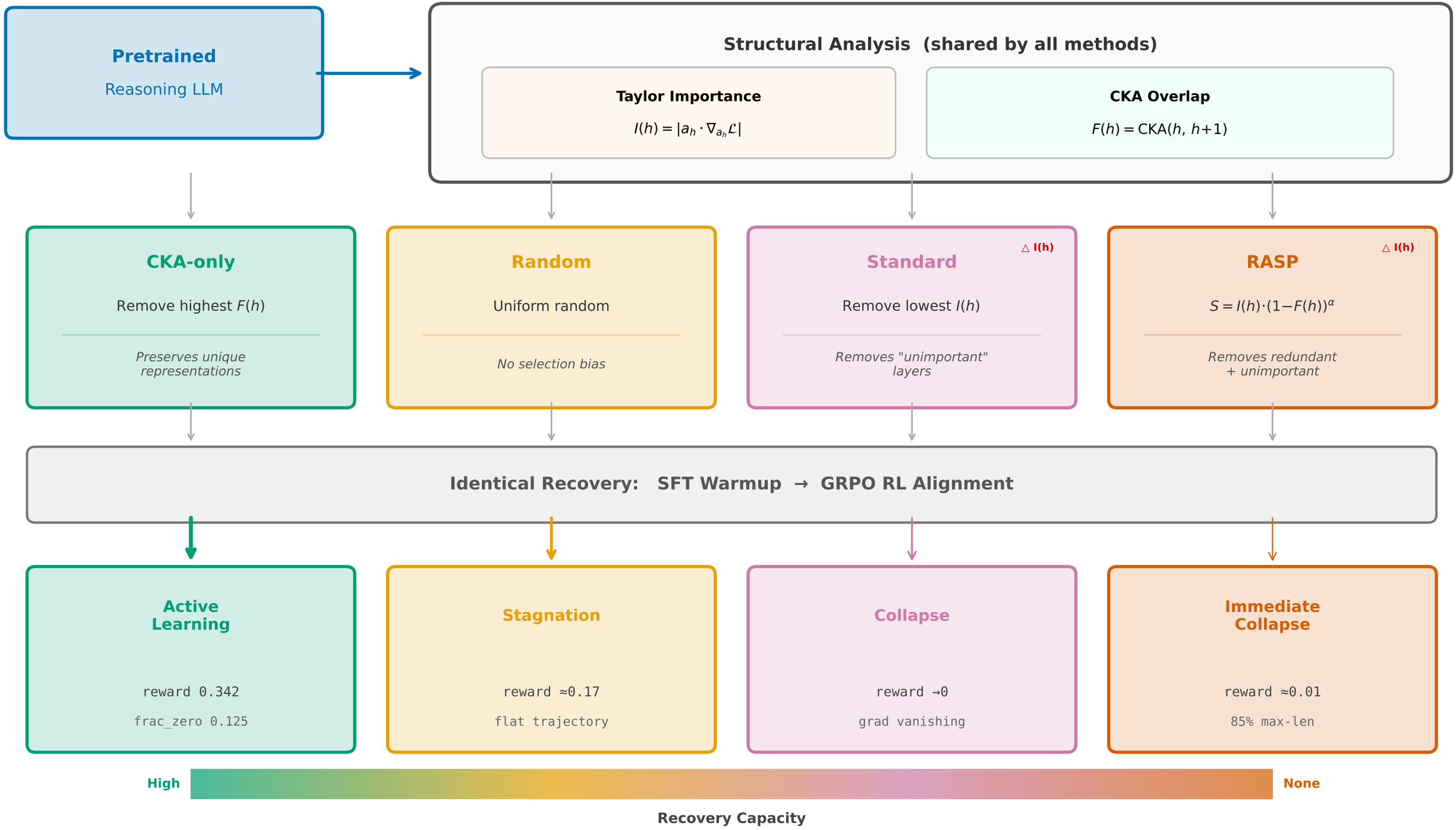

Pretrained
Reasoning LLM
Structural Analysis (shared by all methods)
Taylor Importance
$I(h) = |a_h \cdot \nabla_{a_h} \mathcal{L}|$
CKA Overlap
$F(h) = \mathrm{CKA}(h, h+1)$
CKA-only
Remove highest $F(h)$
Preserves unique representations
Random
Uniform random
No selection bias
Standard
△ I(h)
Remove lowest $I(h)$
Removes "unimportant" layers
RASP
△ I(h)
$S = I(h) \cdot (1 - F(h))^{\alpha}$
Removes redundant + unimportant
Identical Recovery: SFT Warmup → GRPO RL Alignment
Active Learning
reward 0.342
frac_zero 0.125
Stagnation
reward ≈0.17
flat trajectory
Collapse
reward →0
grad vanishing
Immediate Collapse
reward ≈0.01
85% max-len
High
None
Recovery Capacity